

The 22nd International Conference on Practical applications of Agents and Multi-Agent Systems

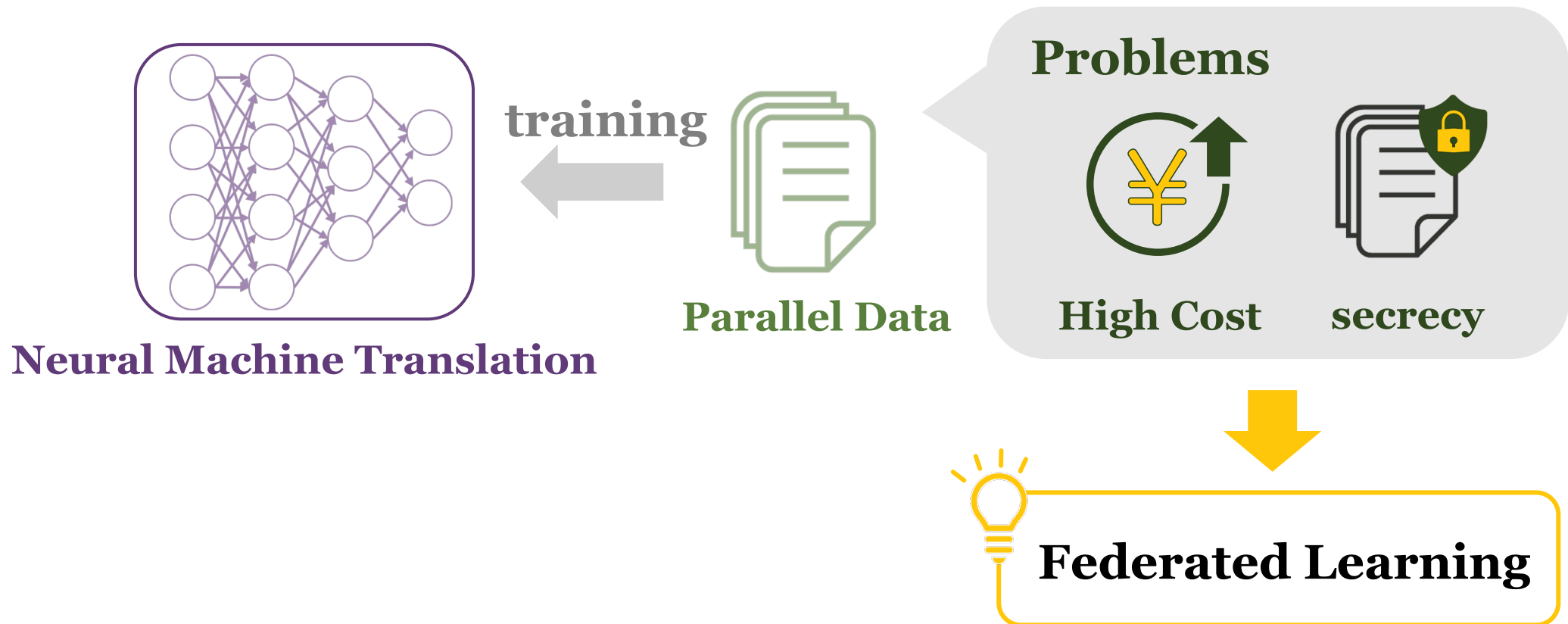
Paper ID : 2928

Federated Neural Machine Translation Using Multi-Agent Reinforcement Learning

Kantaro Kitagawa and Yohei Murakami
Faculty of Information Science and Engineering
Ritsumeikan University

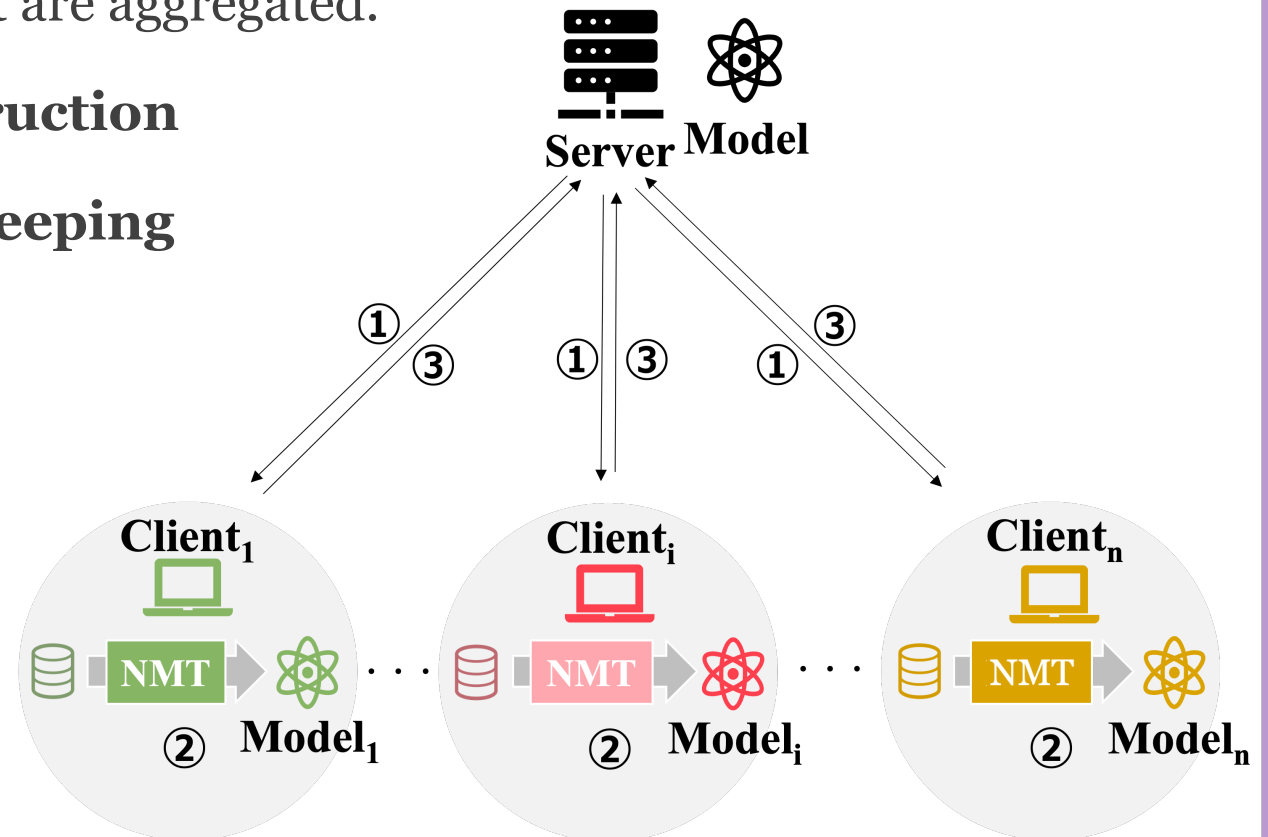
Research Background

- Neural Machine Translation (NMT) has improved the accuracy of machine translation.

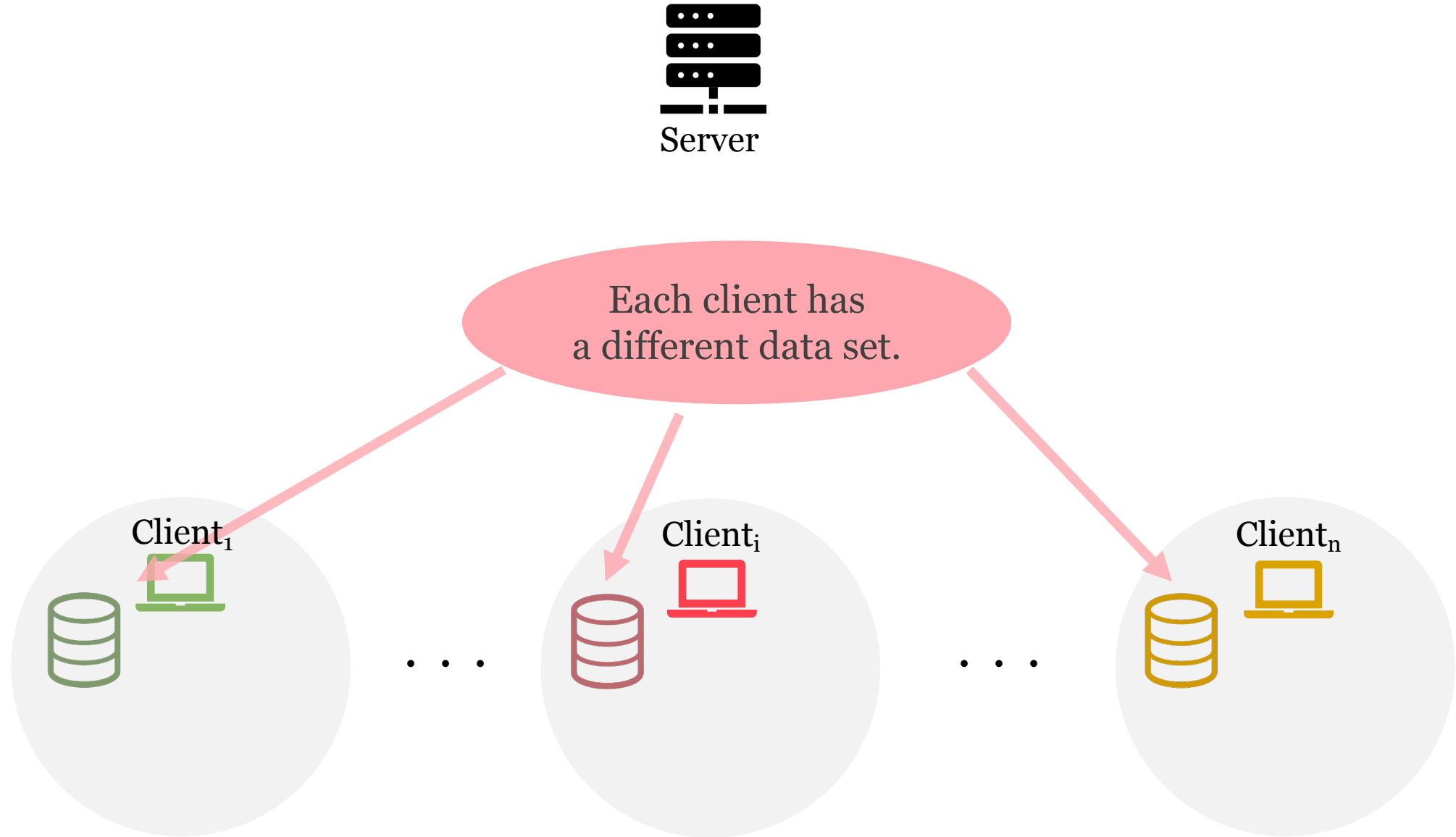


Research Background

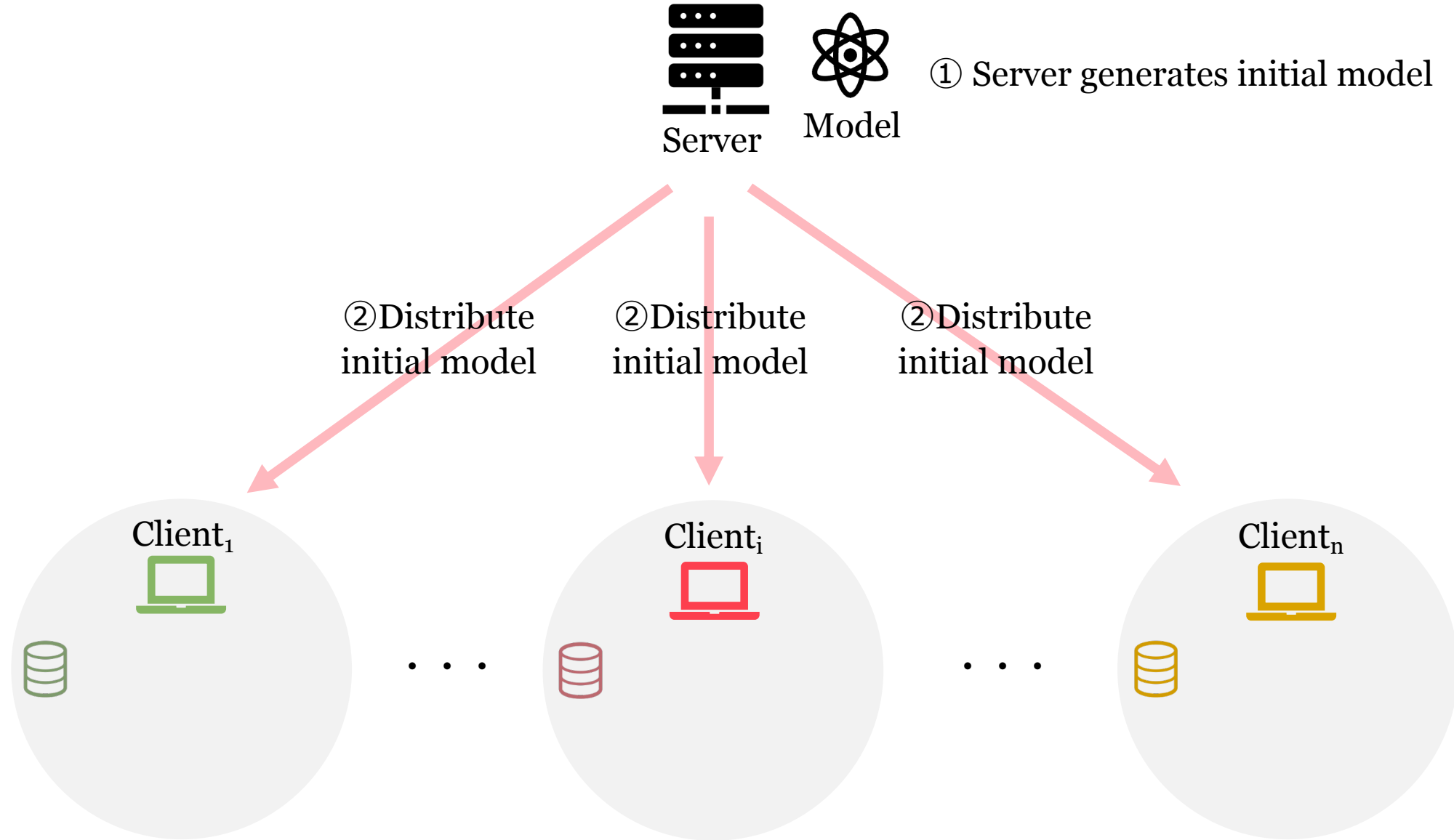
- **Federated learning** is a machine learning technique that does **not** involve data aggregation
 - **Only models** learned by each client are aggregated.
 - Allow for the **collaborative construction** of a machine learning model while **keeping each client's data closed**.



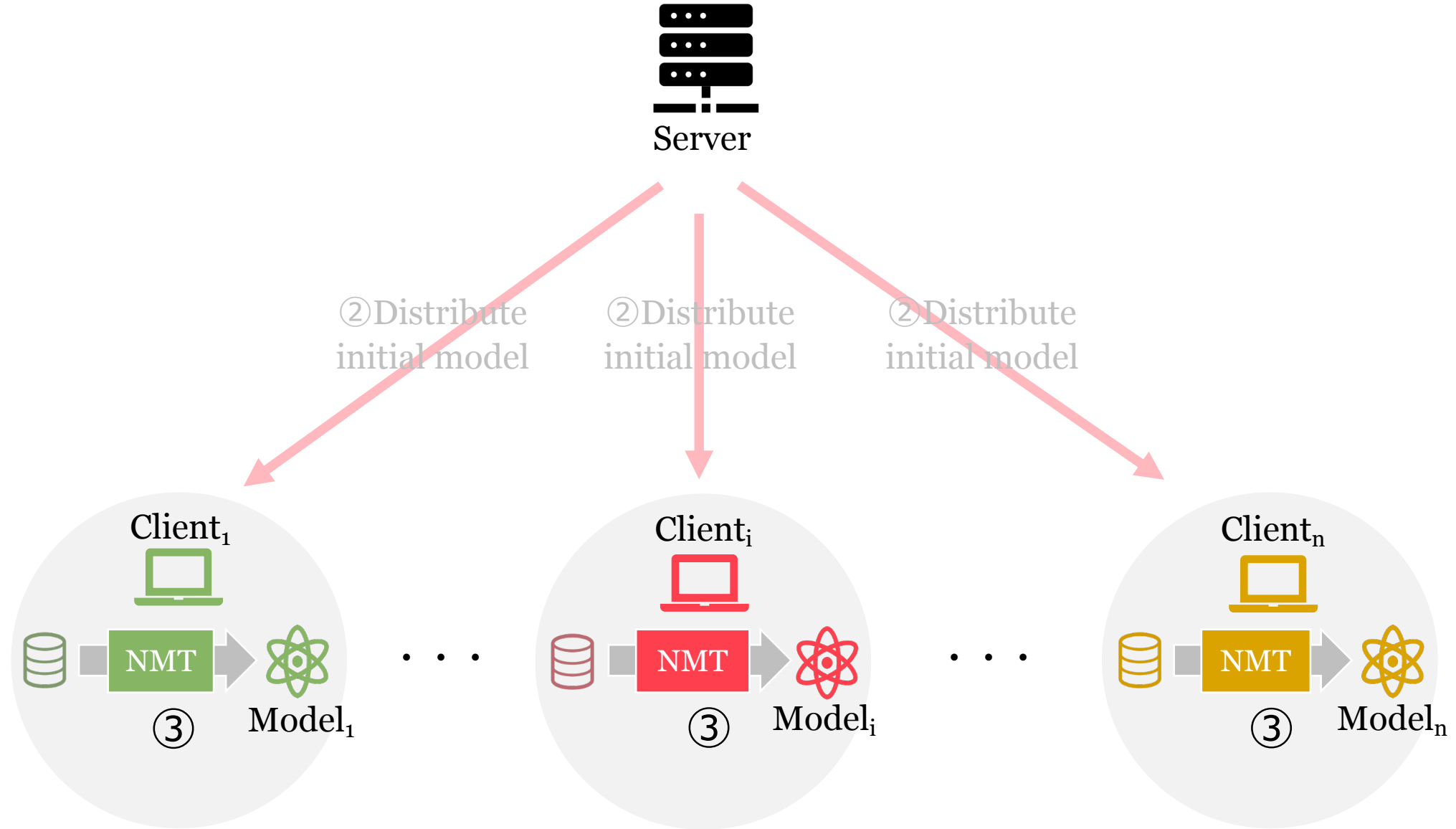
Federated Learning(FedAvg)



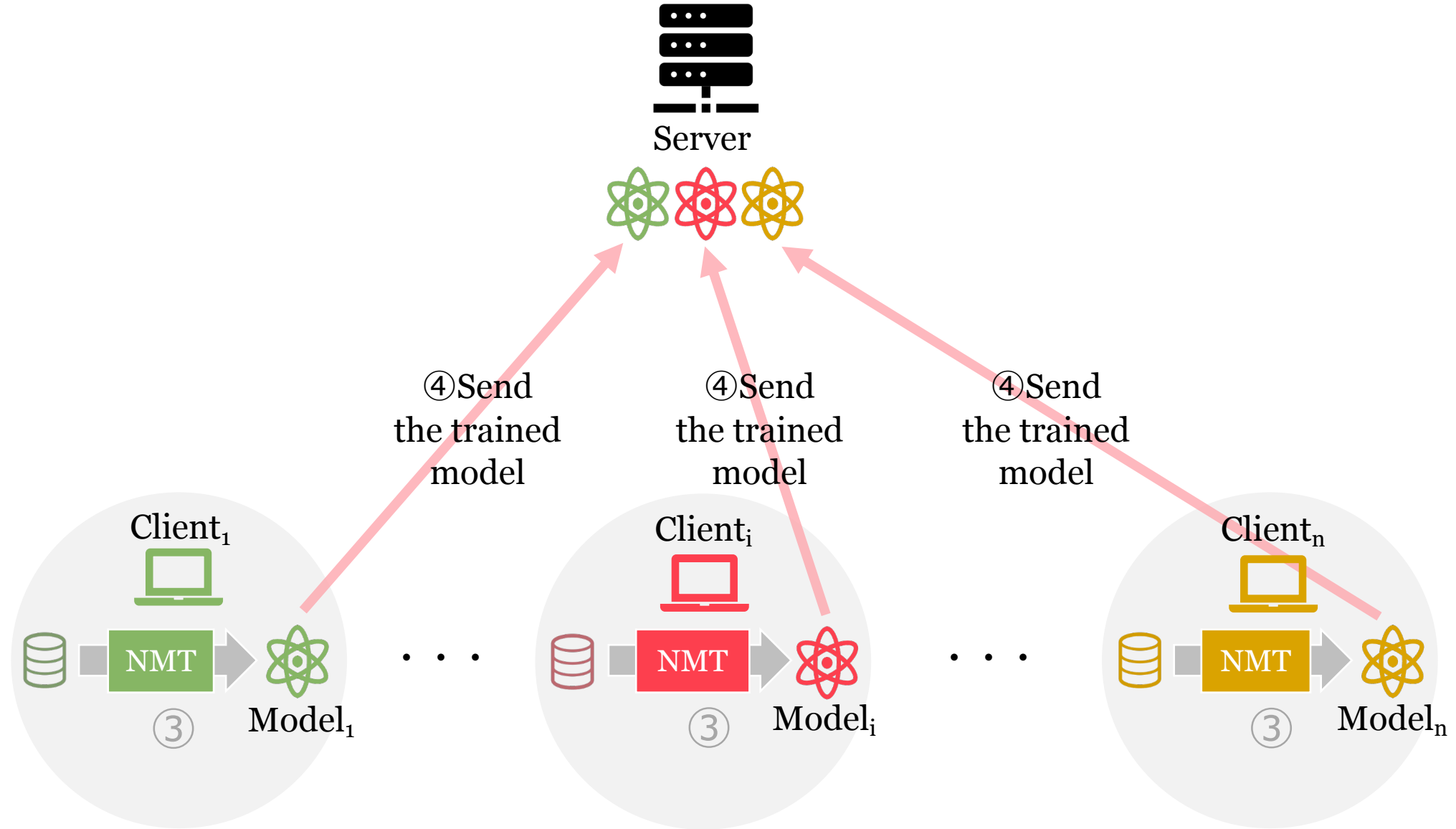
Federated Learning(FedAvg)



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Server

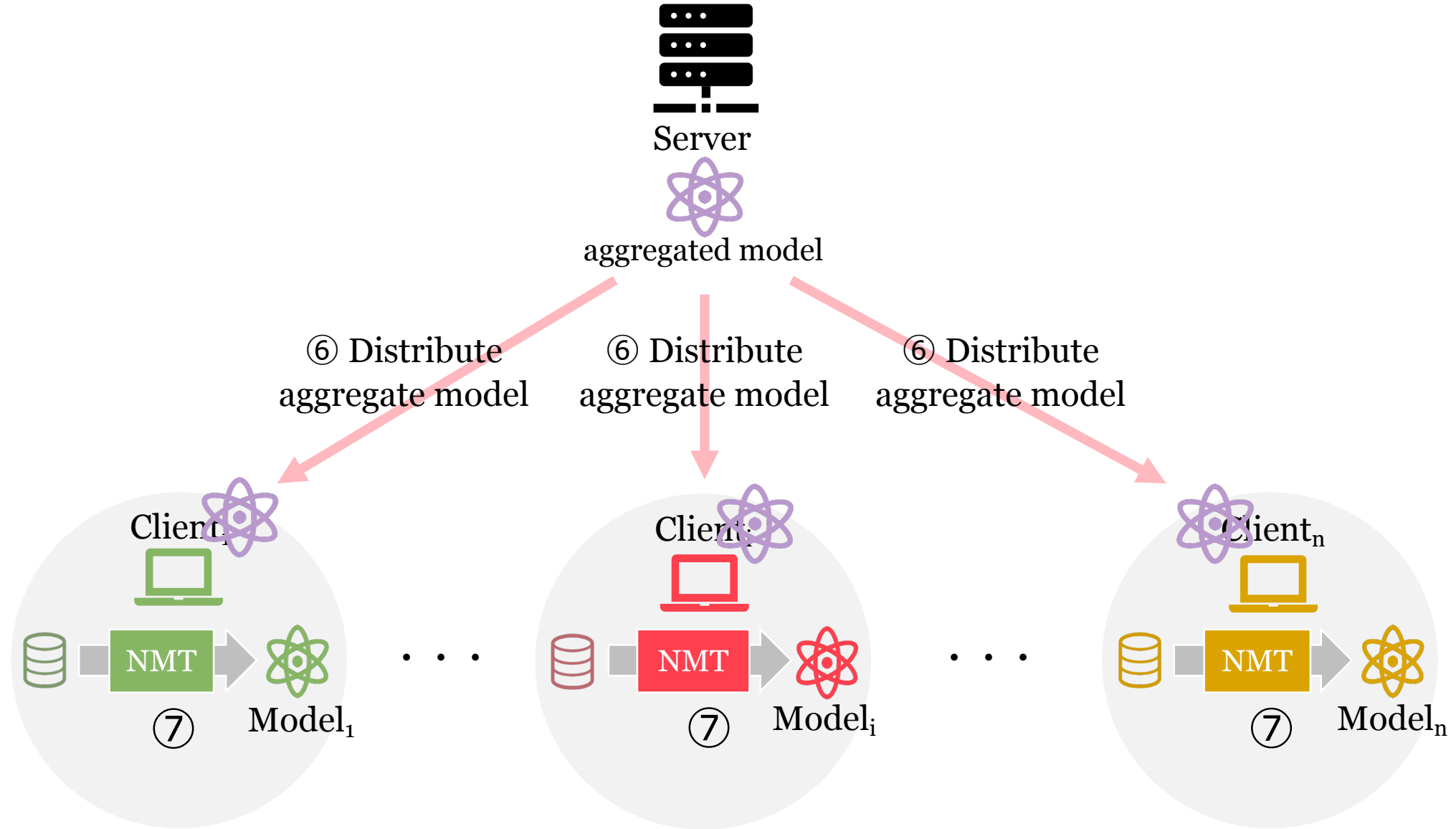


aggregated model

⑤ Server aggregates the model

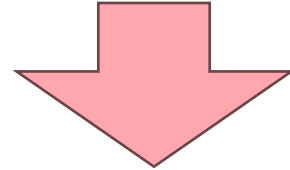


Federated Learning(FedAvg)



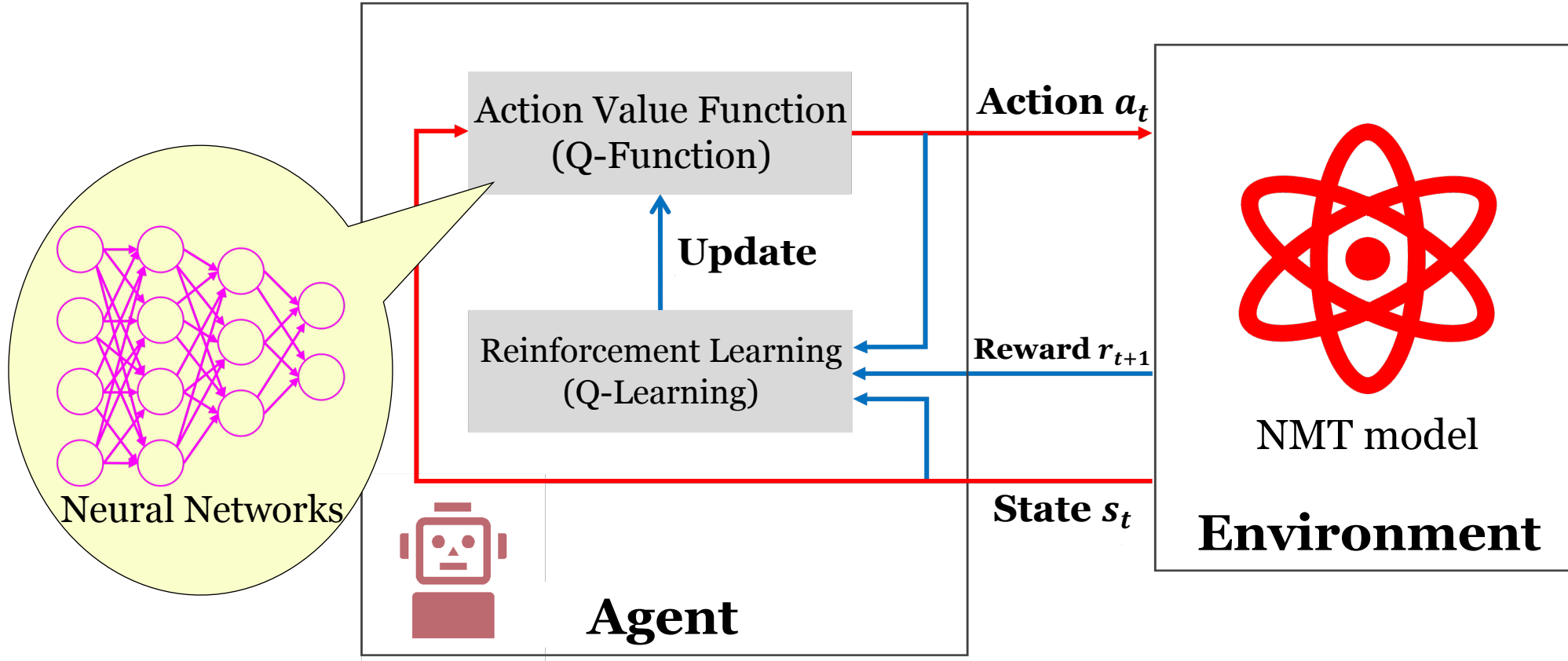
Research Objectives

- Impossible to reflect the preferences of each client during model aggregation.
 - In **non-IID** (non-independent and identically distributed) environments, aggregating all the models may **degrade their translation accuracy**.
 - Each client needs to decide which model to aggregate.

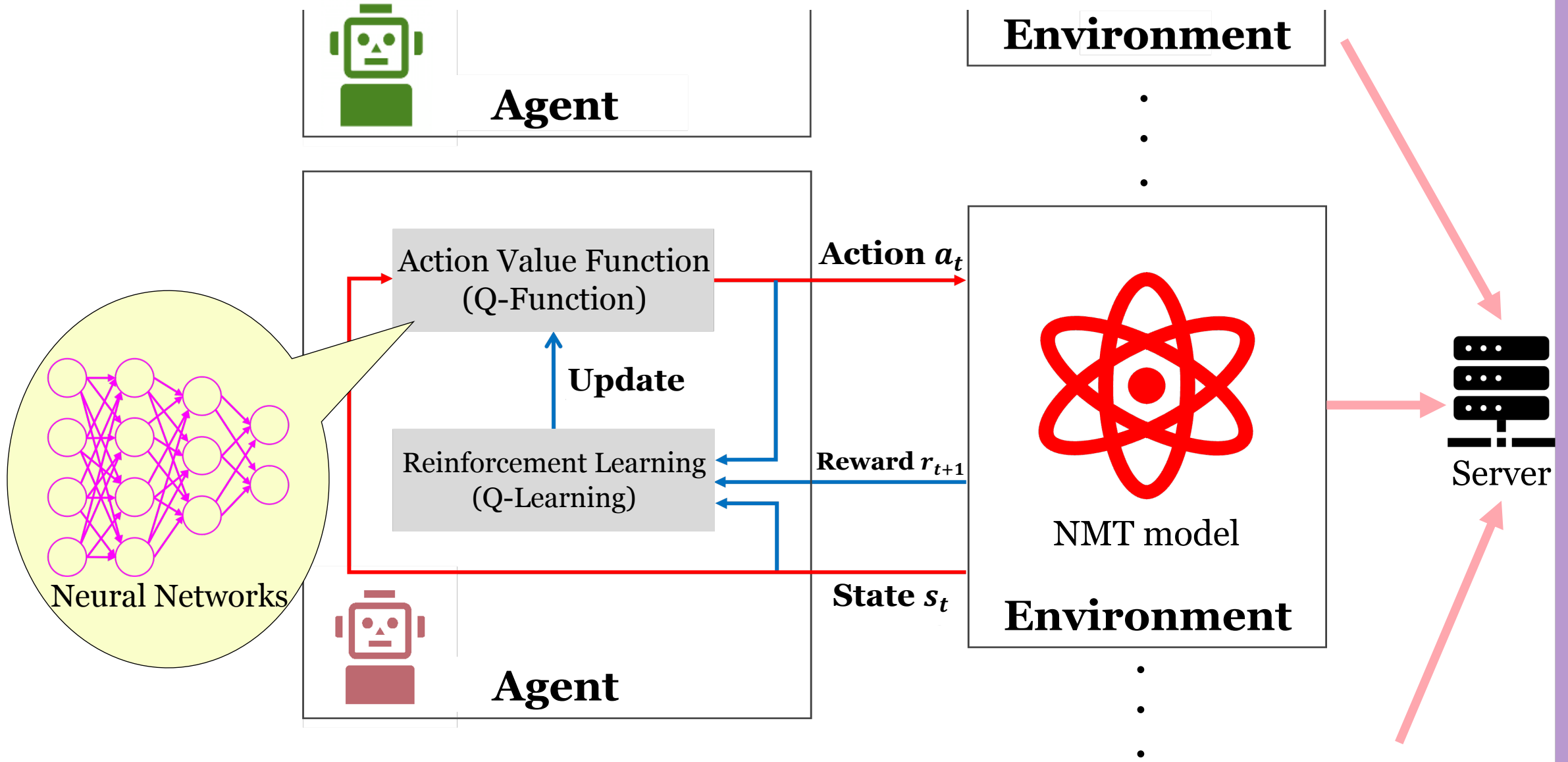


- Propose a method for aggregating models according to the preferences of each client.
 - Acquire a policy using **deep reinforcement learning**.
 - **Individual policies** tailored to each client.

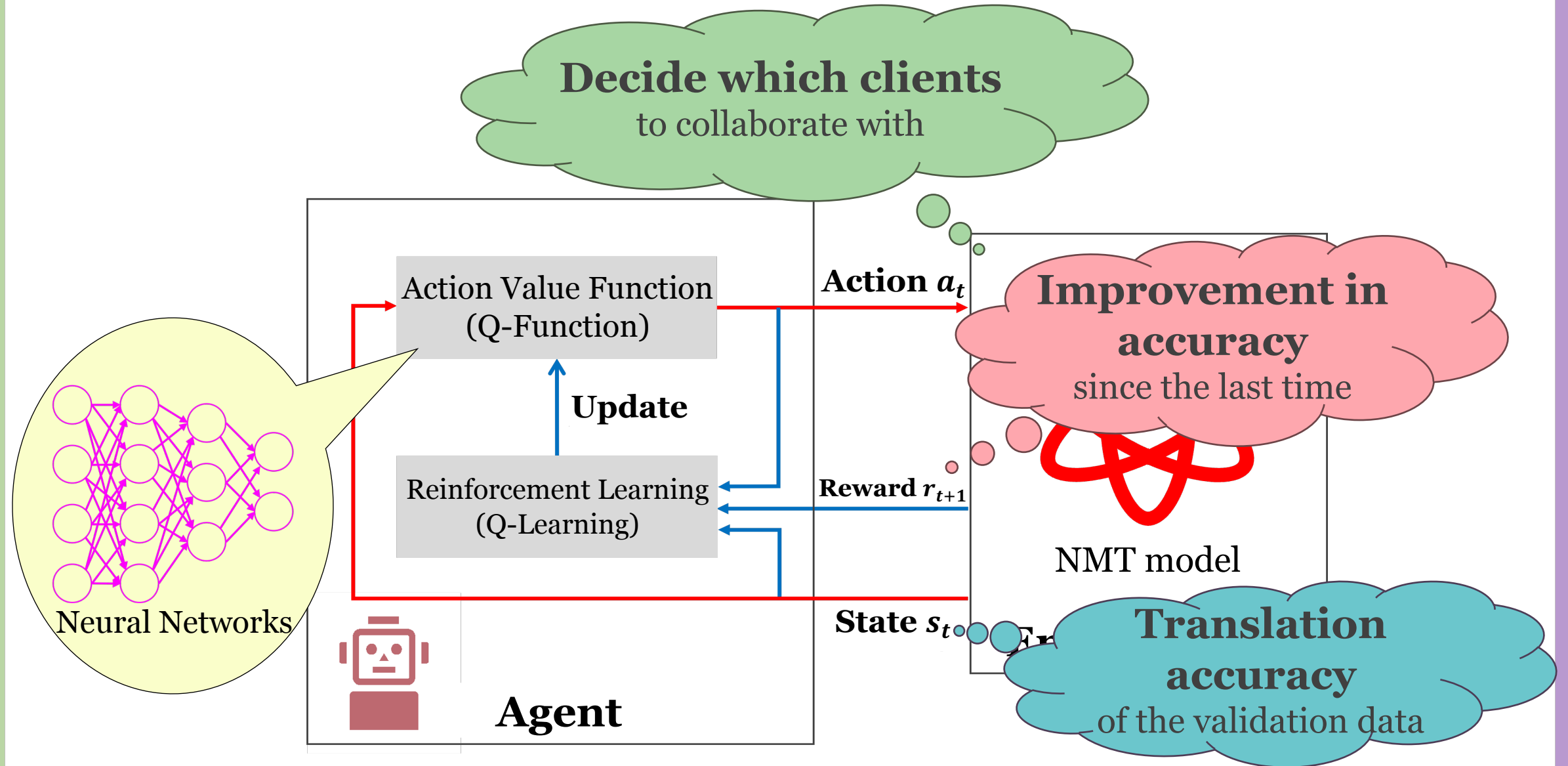
Deep Reinforcement Learning



Deep Reinforcement Learning



Deep Reinforcement Learning



State

- **Definition:**

State s_t is the vector of the translation accuracy of **NMT** at time t

- Accuracy Evaluation Metric: **BLEU score** (using validation data)

- **Validation Data:** Translation pairs(source text st and target text tt)

$$s_t = [BLEU(NMT_t(st_1), tt_1), BLEU(NMT_t(st_2), tt_2), \dots, BLEU(NMT_t(st_k), tt_k)]$$

- BLEU Score Basis: n-gram (bigram) matching between the translated sentence and the correct translation.

Reward

- **Definition:**

Reward r_t is the improvement in translation accuracy since time $t-1$

$$r = score_t - score_{t-1}$$

$$score_t = Average \begin{pmatrix} BLEU(NMT_t(st_1), tt_1), \\ BLEU(NMT_t(st_2), tt_2), \\ \ddots, \\ BLEU(NMT_t(st_k), tt_k) \end{pmatrix}$$

Action

- **Definition:**

Action a is a choice of clients to collaborate with.

Action set A is the power set of the client set CL , excluding the empty set $\emptyset$.

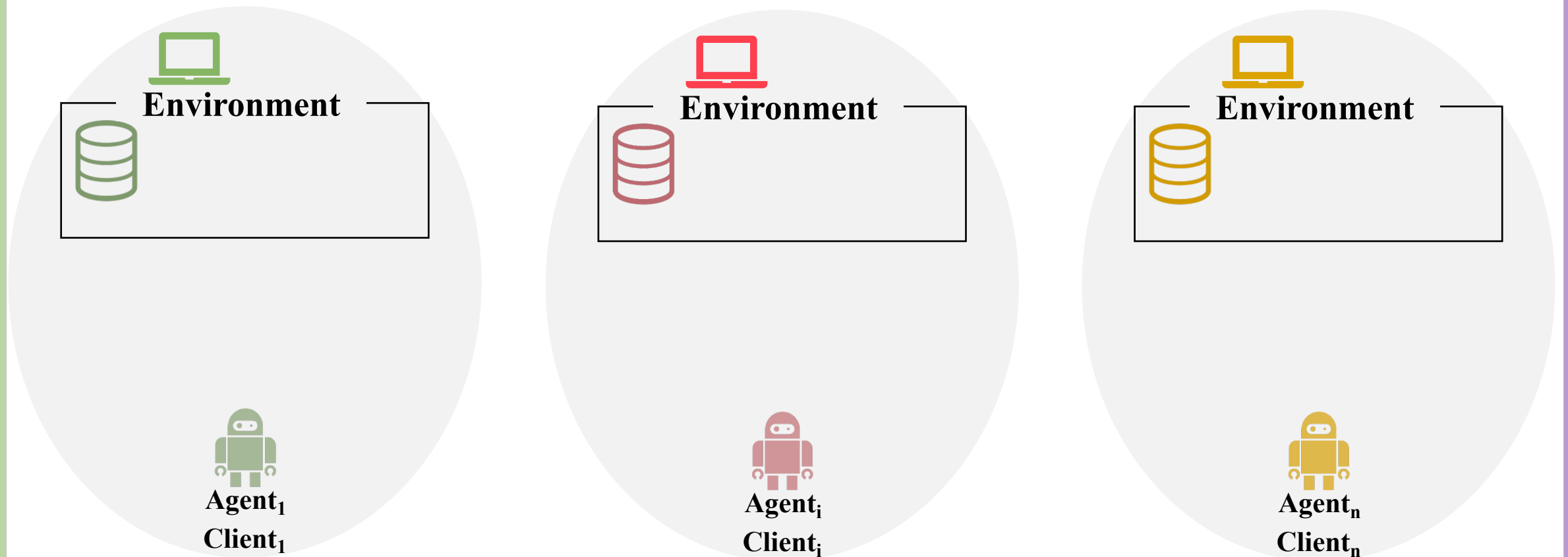
$$A: \mathcal{P}(CL) - \emptyset$$

- Client set CL is as follows.

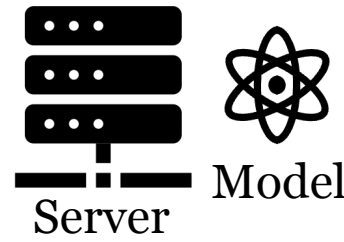
$$CL : \{cl_1, cl_2, \dots, cl_i, \dots, cl_k\}$$

- Action includes the process of aggregating the models of the selected clients and taking the average of the selected models.

MARL-Based FL



MARL-Based FL

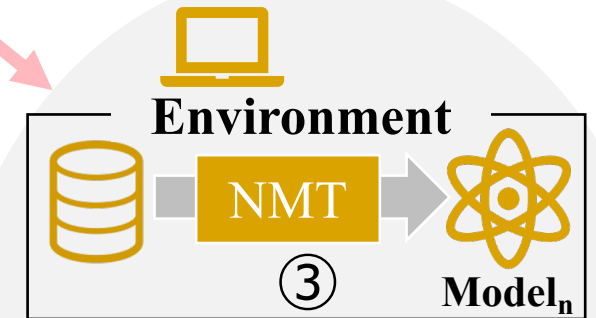
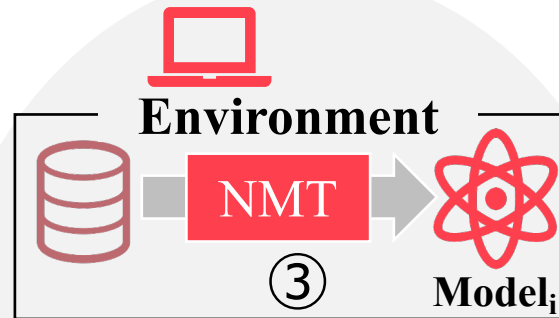
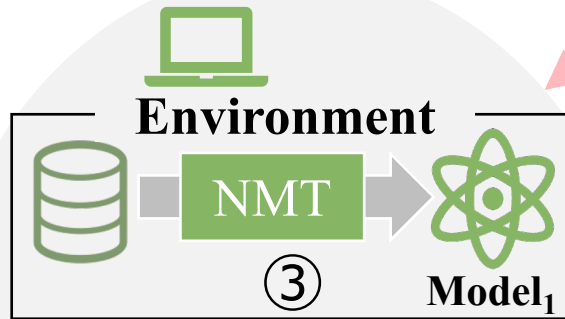


① Server generates initial model

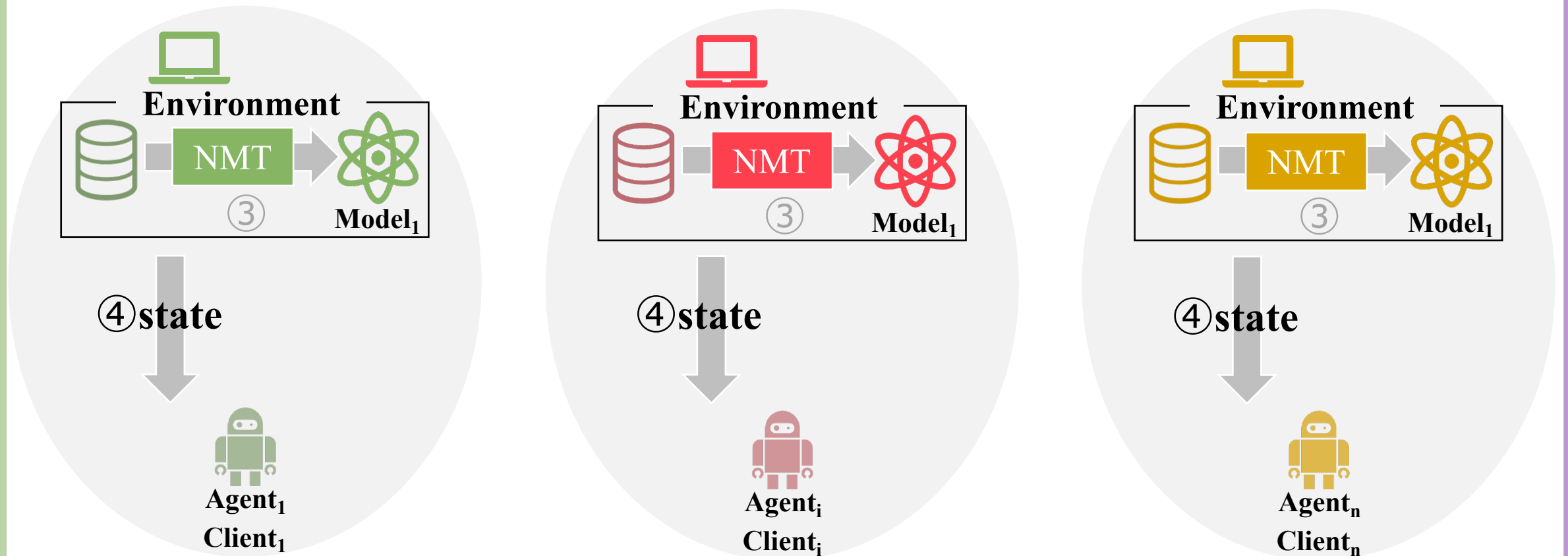
② Distribute
initial model

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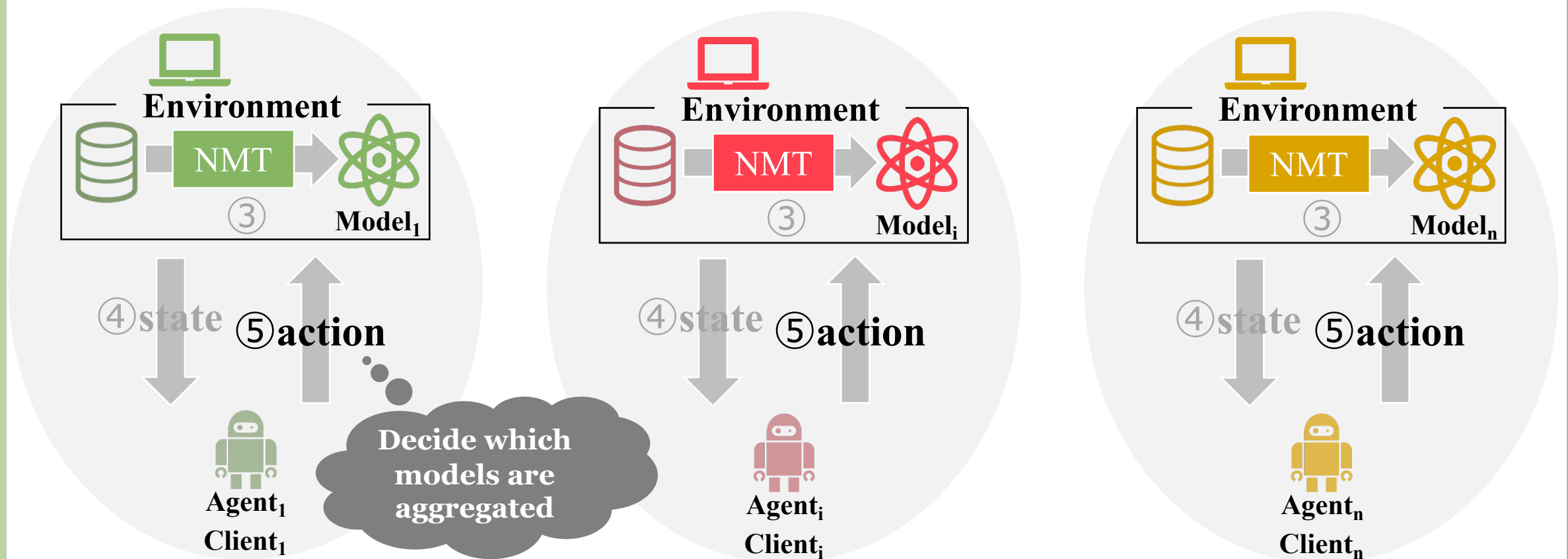
② Distribute
initial model



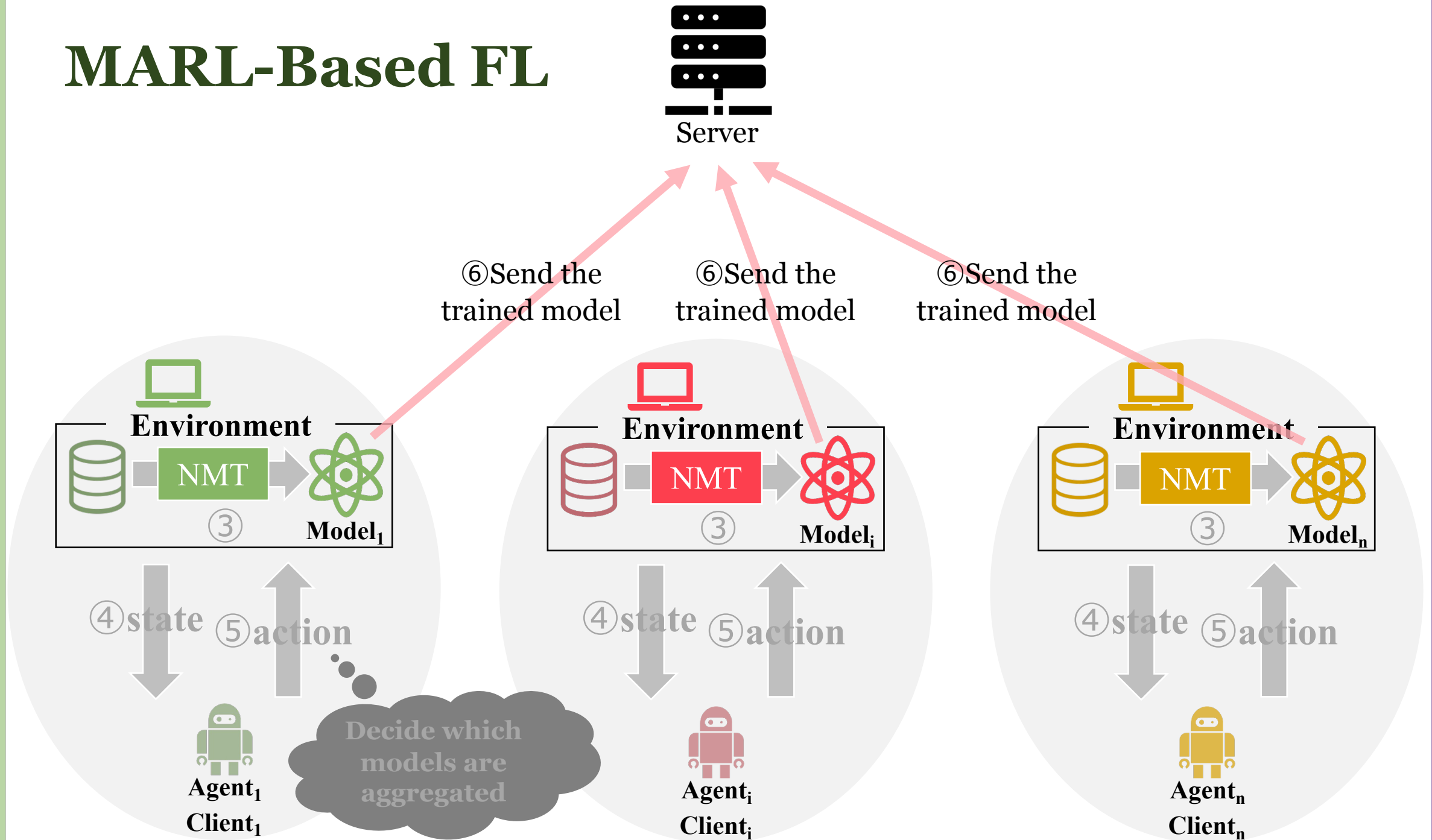
MARL-Based FL



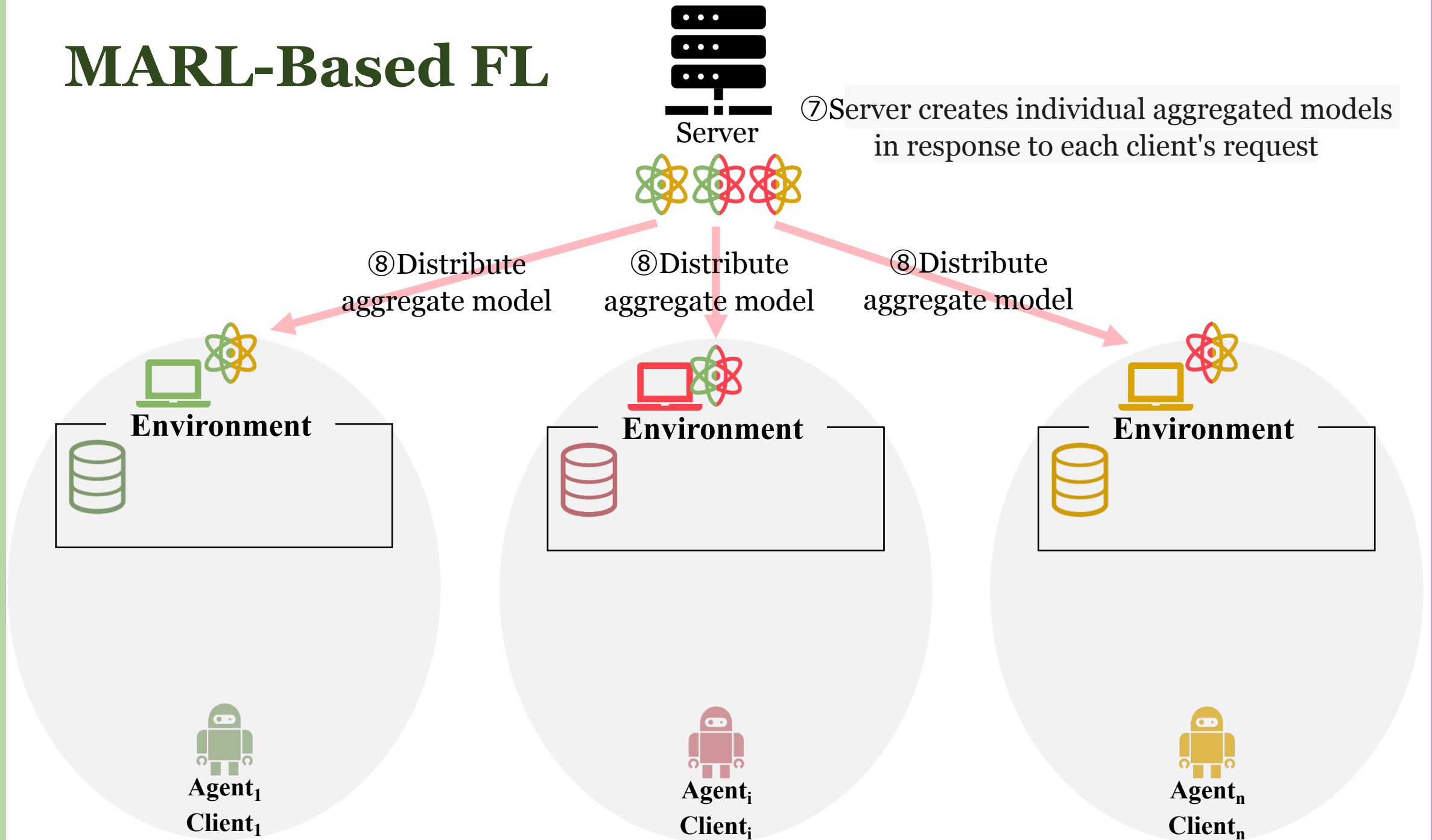
MARL-Based FL



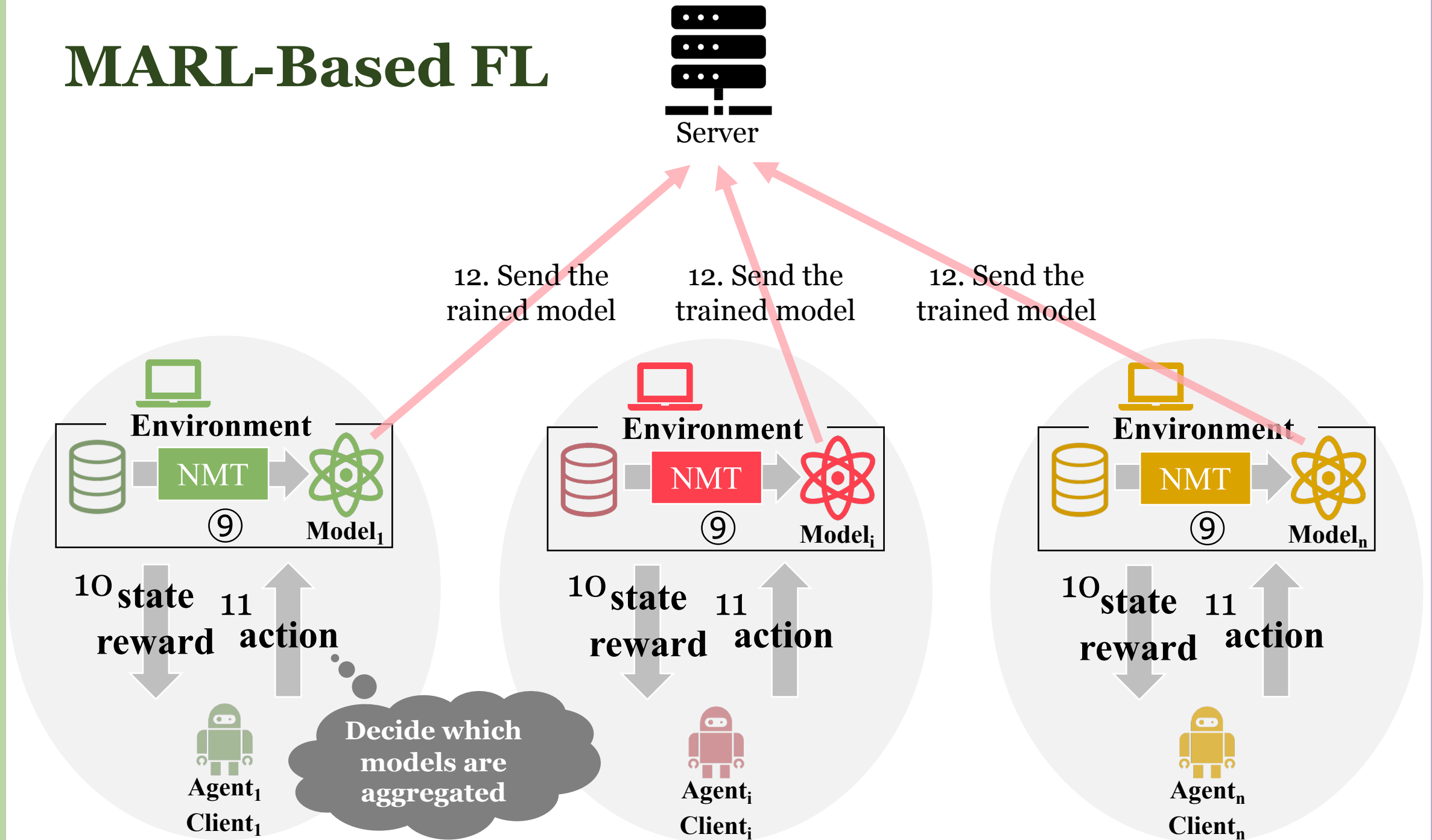
MARL-Based FL



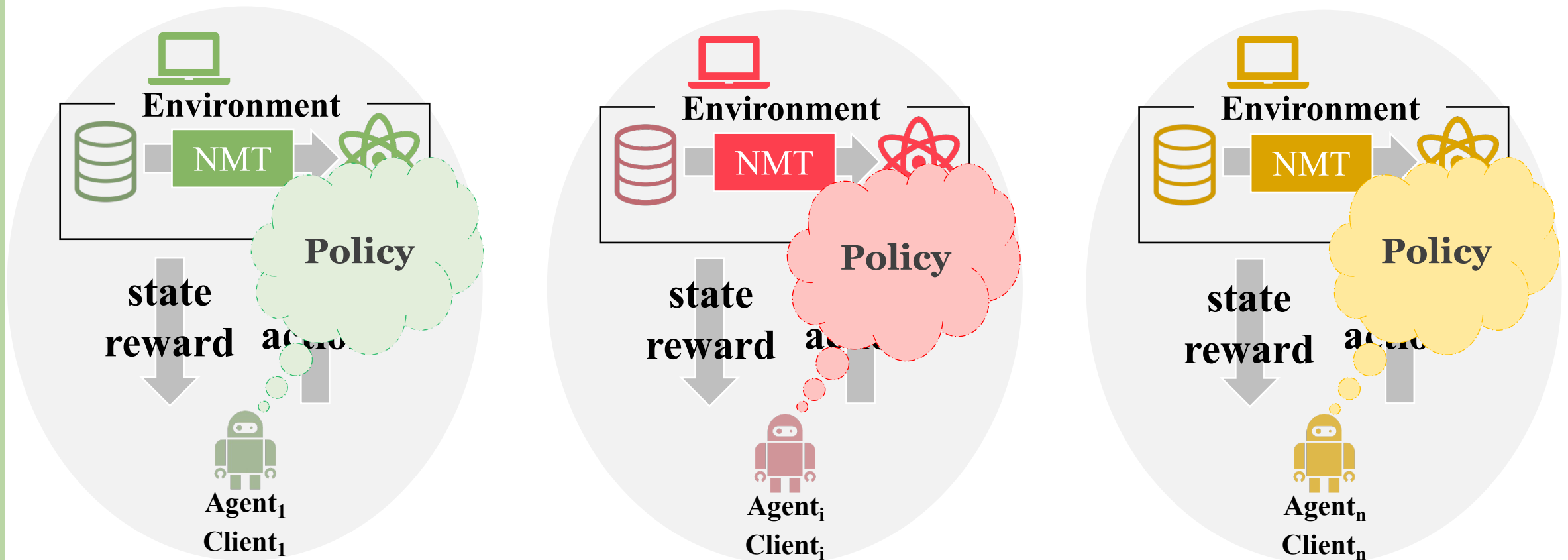
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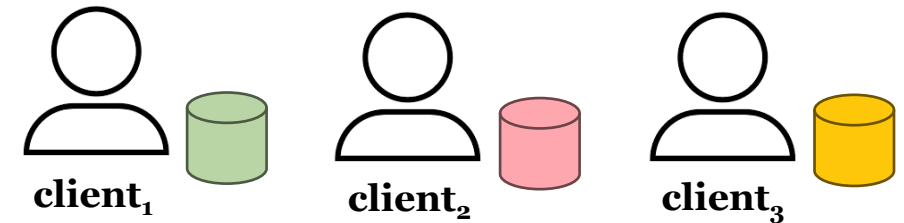
MARL-Based FL



Agent Model

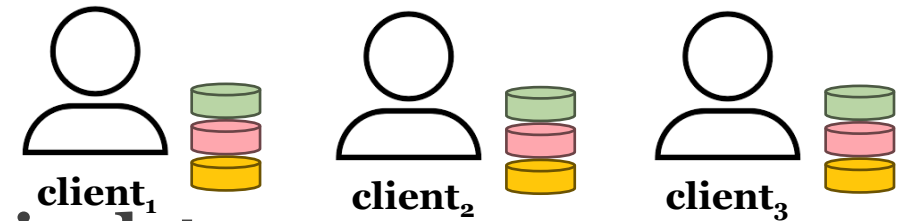
- Agent models, such as state and reward depend on validation data to calculate BLEU score.

- Competitive Agents**



- Validation data: **Individual domain-specific data**
- Prefer to improve translation accuracy in each domain

- Cooperative Agents**

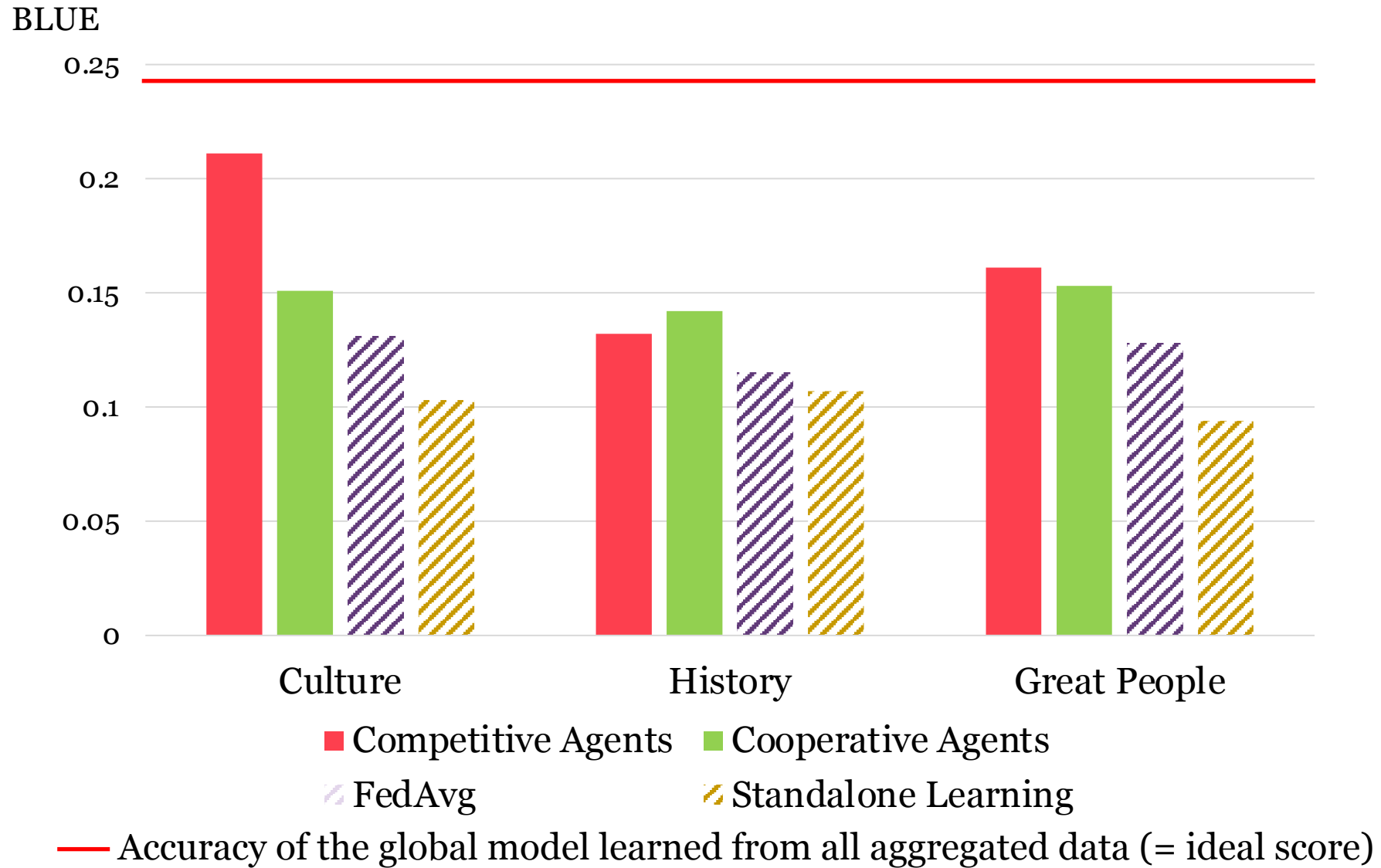


- Validation data: **The same mixed-domain data**
- Prefer to improve translation accuracy in all the domain

Experiment

- Neural machine translation system: "**OpenNMT**"
- Data: **Wikipedia Japanese-English Kyoto Related Documents Parallel Corpus**
- Experimental Environment
 - Number of clients: **3**
 - Domain: "**Culture**," "**History**," and "**Great People**"
 - Data: 80,000 translation pairs in each domain
 - Number of aggregation rounds: **5**
 - Number of reinforcement learning episodes: **460**

Experimental Results



Experimental Results

C: Culture, H: History, G: Great people

aggregation rounds	1st	2nd	3rd	4th	5th
Competitive Agents					
Culture	C + G	C + G	C + G	C + G	C + G
History	C + H + G	C + G	C + H + G	C	C + G
Great People	C + G	C + G	C + G	C + G	C + G
Cooperative Agents					
Culture	C	C	C	C + H + G	H + G
History	C + H	C	C	C	C
Great People	C	C	C	C	C

- We traced the models used by each agent and analyzed the connections between the agents.
- In competitive agents, "C + G" was used for the "Culture" and "Great People", improving the accuracy of each.

Discussion

- We calculated the similarity of each corpus using the LDA topic model.

Topic 1: Japanese Culture and Heritage

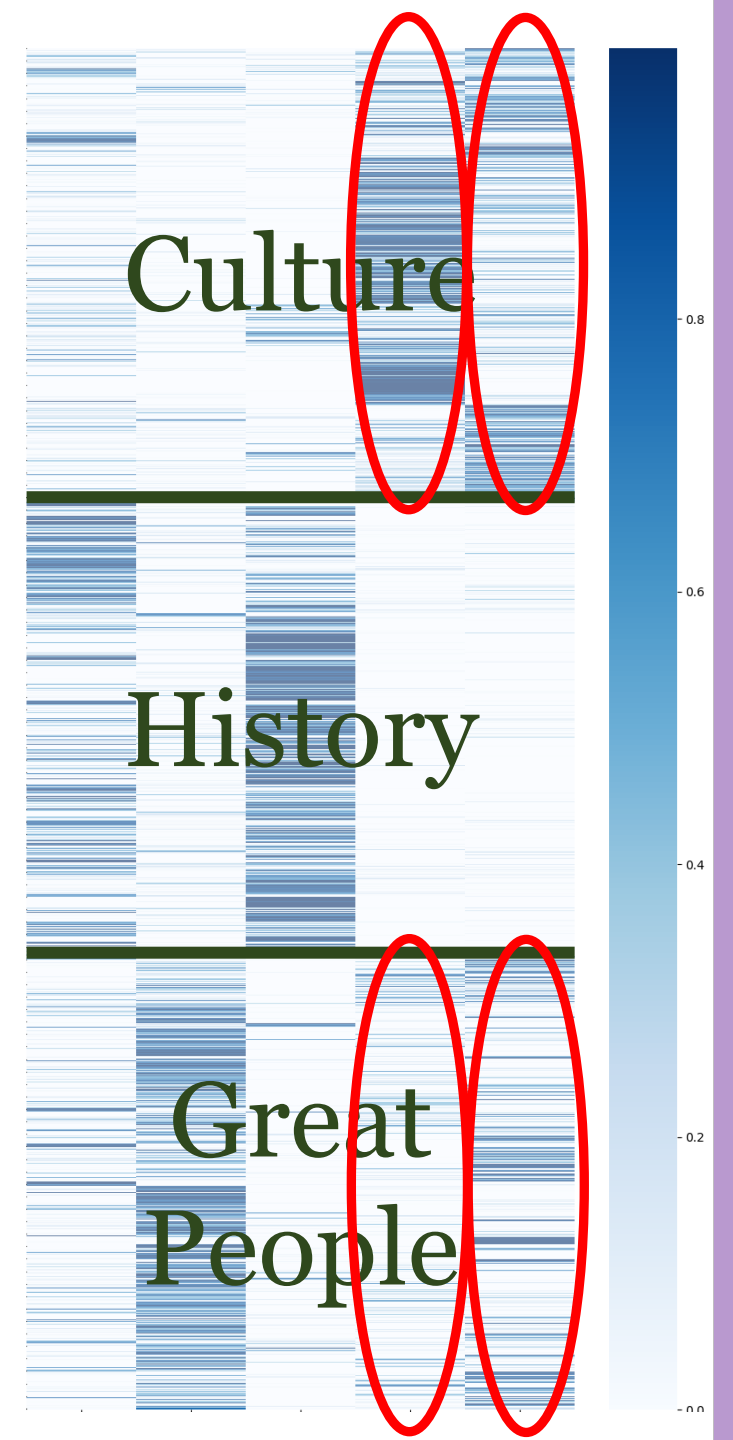
Topic 2: Japanese History and Government

Topic 3: Japanese Cooking and Food Culture

Topic 4: **Japanese Royalty and Nobles**

Topic 5: **Japanese Sengoku period**

- Cosine similarity of each document:
 - Culture and History: 0.22
 - Culture and Great People: **0.51**
 - History and Great People : 0.24



Discussion

- Training data are **partially overlapped**.
- Improvements in one domain elevates the overlapping areas.
- **Other domains** are also improved.
- In a federated learning environment, the behavior of competitive agents is selfish, but it becomes indirectly **cooperative** because federated learning is originally cooperative framework.

Conclusion

- Propose a method to **dynamically select collaborators** during each aggregation process by incorporating **deep reinforcement learning** into **federated learning**.
- Demonstrate higher accuracy compared to the traditional FedAvg method, proving its effectiveness.
- **Competitive agents** were able to enhance the specialization of the neural machine translation model and significantly improve its accuracy.